

# Assignment 1: Java program and collaboration

CS3354

This is a group assignment.

You have been randomly assigned to a team on Canvas.

Contact your team members and collaborate to complete this program.

To collaborate with the code, a Github repository must be created and the other team members must be added as collaborators.

## Objective

Collaborate as a team to build a grocery management system using parallel arrays. You will manage data across multiple arrays where the same index refers to the same item.

## Team Roles & Git Workflow

Each student must create their own branch in Git, perform their task, and then merge it into the main branch.

## Tasks

### The Task: Parallel Array Architecture

You will maintain the following data across three arrays in your main method:

- `String[] itemNames = new String[10];`
- `double[] itemPrices = new double[10];`
- `int[] itemStocks = new int[10];`

### Task 1: Inventory Display

**Branch:** feature-display

**Method:** `public static void printInventory(String[] names, double[] prices, int[] stocks)`

**Logic:** Use a for loop to iterate through the arrays.

**Requirement:** Use an if-else statement inside the loop to only print slots that aren't empty (e.g., if `(names[i] != null)`).

### Task 2: Restock & Search

**Branch:** feature-restock

**Method:** `public static void restockItem(String[] names, int[] stocks, String target, int amount)`

**Logic:** Use a loop to find the target name. If found, add the amount to that index in the stocks array.

**Requirement:** If the item isn't found after checking the whole loop, print "Item not found."

## Task 3: The User Menu

**Branch:** feature-menu

**Logic:** In the main method, use a Scanner and a while(true) loop to create a menu.

**Integration:** Call the methods written in the previous two tasks above based on the user's input (1 for View, 2 for Restock, 3 to Exit).

## Additional requirements

### Documentation

The program class must have an appropriate javadoc comment.

Each method in the class must have an appropriate javadoc comment.

Generate the javadoc documentation and store it in a docs/ folder within your project folder.

### Submission

Submit the URL to your GitHub repository.

### Grading

To grade your project, please add Teaching Assistant's email as your collaborator of this project.

### Note

Two (2) more activities related to this assignment will be posted. In these assignments, you will evaluate your peers' collaboration.